Number Theory
CS1010 Discrete Mathematics for Computer Science

Shibam Ghosh

August 31, 2026

Department of CSE, IIT Hyderabad

Greatest Common Divisor (GCD)

Greatest Common Divisor

Definition

Let a and b be integers, not both zero. The largest integer d that divides a and b is called the **greatest common divisor** of a and b, denoted $\gcd(a, b)$.

- $\gcd(32, 24) = 8$
- $\gcd(27, 11) = 1$

Definitions

- The integers a and b are **relatively prime** if $\gcd(a, b) = 1$.
- Integers $a_1, a_2, \ldots, a_n$ are **pairwise relatively prime** if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Greatest Common Divisor

Definition

Let a and b be integers, not both zero. The largest integer d that divides a and b is called the **greatest common divisor** of a and b, denoted $\gcd(a, b)$.

- $\gcd(32, 24) = 8$
- $\gcd(27, 11) = 1$

Definitions

- The integers a and b are **relatively prime** if $\gcd(a, b) = 1$.
- Integers $a_1, a_2, \ldots, a_n$ are **pairwise relatively prime** if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Pairwise Relatively Prime: Examples

- Determine whether 22, 27, and 29 are pairwise relatively prime.
- **Yes:** $\gcd(22, 27) = 1$, $\gcd(22, 29) = 1$, and $\gcd(27, 29) = 1$.
- Determine whether 16, 17, and 24 are pairwise relatively prime.
- **No:** $\gcd(16, 17) = 1$, $\gcd(16, 24) = 8$, and $\gcd(17, 24) = 1$.

Pairwise Relatively Prime: Examples

- Determine whether 22, 27, and 29 are pairwise relatively prime.
- **Yes:** $\gcd(22, 27) = 1, \gcd(22, 29) = 1$, and $\gcd(27, 29) = 1$.
- Determine whether 16, 17, and 24 are pairwise relatively prime.
- **No:** $\gcd(16, 17) = 1, \gcd(16, 24) = 8,$ and $\gcd(17, 24) = 1$.

Finding GCD Using Prime Factorizations

Suppose prime factorizations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$

where exponents are ≥ 0. Then:

$$\gcd(a, b) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} \dots p_n^{\min(a_n,b_n)}$$

Example:

$$96 = 2^5 \cdot 3^1$$

$$196 = 2^2 \cdot 7^2$$

$$\gcd(96, 196) = 2^{\min(5,2)} \cdot 3^{\min(1,0)} \cdot 7^{\min(0,2)} = 2^2 \cdot 3^0 \cdot 7^0 = 4$$

Note: Not efficient for huge numbers since factoring is hard.

Finding GCD Using Prime Factorizations

Suppose prime factorizations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$

where exponents are ≥ 0. Then:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \dots p_n^{\min(a_n, b_n)}$$

Example:

$$96 = 2^5 \cdot 3^1$$
$$196 = 2^2 \cdot 7^2$$
$$\gcd(96, 196) = 2^{\min(5,2)} \cdot 3^{\min(1,0)} \cdot 7^{\min(0,2)} = 2^2 \cdot 3^0 \cdot 7^0 = 4$$

Note: Not efficient for huge numbers since factoring is hard.

Finding GCD Using Prime Factorizations

Suppose prime factorizations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$

where exponents are ≥ 0. Then:

$$\gcd(a, b) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} \dots p_n^{\min(a_n,b_n)}$$

Example:

$$\begin{aligned} 96 &= 2^5 \cdot 3^1 \\ 196 &= 2^2 \cdot 7^2 \\ \gcd(96, 196) &= 2^{\min(5,2)} \cdot 3^{\min(1,0)} \cdot 7^{\min(0,2)} = 2^2 \cdot 3^0 \cdot 7^0 = 4 \end{aligned}$$

Note: Not efficient for huge numbers since factoring is hard.

The Euclidean Algorithm

Key Idea

An efficient method for computing $\gcd(a, b)$ based on the fact:

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

Show that $\gcd(414, 162) = 18$:

$$\begin{aligned}
414 &= 162 \cdot 2 + 90 && \implies \gcd(414, 162) = \gcd(162, 90) \\
162 &= 90 \cdot 1 + 72 && \implies \gcd(162, 90) = \gcd(90, 72) \\
90 &= 72 \cdot 1 + 18 && \implies \gcd(90, 72) = \gcd(72, 18) \\
72 &= 18 \cdot 4 + 0 && \implies \text{Stop. Remainder is 0.}
\end{aligned}$$

The Euclidean Algorithm

Key Idea

An efficient method for computing $\gcd(a, b)$ based on the fact:

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

Show that $\gcd(414, 162) = 18$:

$$\begin{aligned}
414 &= 162 \cdot 2 + 90 && \implies \gcd(414, 162) = \gcd(162, 90) \\
162 &= 90 \cdot 1 + 72 && \implies \gcd(162, 90) = \gcd(90, 72) \\
90 &= 72 \cdot 1 + 18 && \implies \gcd(90, 72) = \gcd(72, 18) \\
72 &= 18 \cdot 4 + 0 && \implies \text{Stop. Remainder is 0.}
\end{aligned}$$

The Euclidean Algorithm in Action

The Algorithm to find $\gcd(a, b)$:

1. $x := a$
2. $y := b$
3. While $y \neq 0$ holds:
4. $\quad r := x \bmod y$
5. $\quad x := y$
6. $\quad y := r$
7. Return x

Watch how the variables shift color from one line to the next row!

Example: $\gcd(726, 222)$

Tracing values at the start of each step:

x: blue, y: orange

Step 1: $726 = 222 \cdot 3 + 60$

Step 2: $222 = 60 \cdot 3 + 42$

Step 3: $60 = 42 \cdot 1 + 18$

Step 4: $42 = 18 \cdot 2 + 6$

Step 5: $18 = 6 \cdot 3 + \mathbf{0}$

Next step would have $y = 0$, so the loop terminates. The last active x value is our answer.

$$\therefore \gcd(726, 222) = 6$$

Correctness of Euclidean Algorithm (1)

lemma

Let $r \equiv a \bmod b$. Then $\gcd(a, b) = \gcd(b, r)$.

Proof:

- As $r \equiv a \bmod b$, there exists an integer q such that $a = bq + r$.
- Suppose $d \mid a$ and $d \mid b$. Then d also divides $a - bq = r$. Hence, any common divisor of a and b is a common divisor of b and r.
- Conversely, suppose $d \mid b$ and $d \mid r$. Then d also divides $bq + r = a$. Hence, any common divisor of b and r is a common divisor of a and b.
- Therefore, $\gcd(a, b) = \gcd(b, r)$. $\square$

Correctness of Euclidean Algorithm (2)

Let $r_0 = a$ and $r_1 = b$. Successive applications of the division algorithm yield:

$$\begin{aligned} r_0 &= r_1 q_1 + r_2 \quad (0 \le r_2 < r_1) \\ r_1 &= r_2 q_2 + r_3 \quad (0 \le r_3 < r_2) \\ &\vdots \\ r_{n-2} &= r_{n-1} q_{n-1} + r_n \quad (0 \le r_n < r_{n-1}) \\ r_{n-1} &= r_n q_n + 0 \end{aligned}$$

By the Lemma: $\gcd(a, b) = \gcd(r_0, r_1) = \cdots = \gcd(r_{n-1}, r_n) = \gcd(r_n, 0) = r_n$

Hence, the GCD is the **last nonzero remainder.**

GCSs as Linear Combinations

Bézout's Identity

If a and b are positive integers, there exist integers s and t such that:

$$\gcd(a, b) = sa + tb \quad (\textit{Bézout coefficients: } s, t)$$

Step 1: Forward Path

Run the Euclidean Algorithm $\gcd(52, 198)$:

$$198 = 52 \cdot 3 + \mathbf{42} \implies \mathbf{42} = 198 - 52 \cdot 3$$
$$52 = \mathbf{42} \cdot 1 + \mathbf{10} \implies \mathbf{10} = 52 - \mathbf{42} \cdot 1$$
$$\mathbf{42} = \mathbf{10} \cdot 4 + \mathbf{2} \implies \mathbf{2} = \mathbf{42} - \mathbf{10} \cdot 4$$
$$\mathbf{10} = \mathbf{2} \cdot 5 + 0 \quad \text{(Stop)}$$

Step 2: Backward Path

$$2 = \mathbf{42} - \mathbf{10} \cdot 4$$
$$2 = \mathbf{42} - (52 - \mathbf{42} \cdot 1) \cdot 4$$
$$2 = \mathbf{42} \cdot 5 - 52 \cdot 4$$
$$2 = (198 - 52 \cdot 3) \cdot 5 - 52 \cdot 4$$
$$2 = 198 \cdot 5 - 52 \cdot 15 - 52 \cdot 4$$
$$\mathbf{2} = \mathbf{198} \cdot (5) + \mathbf{52} \cdot (-19)$$

Bézout coefficients: $\mathbf{s} = -\mathbf{19}$ and $\mathbf{t} = \mathbf{5}$

GCSs as Linear Combinations

Bézout's Identity

If a and b are positive integers, there exist integers s and t such that:

$$\gcd(a, b) = sa + tb \quad (\textit{Bézout coefficients: } s, t)$$

Step 1: Forward Path

Run the Euclidean Algorithm $\gcd(52, 198)$:

$$198 = 52 \cdot 3 + \mathbf{42} \implies \mathbf{42} = 198 - 52 \cdot 3$$
$$52 = \mathbf{42} \cdot 1 + \mathbf{10} \implies \mathbf{10} = 52 - \mathbf{42} \cdot 1$$
$$\mathbf{42} = \mathbf{10} \cdot 4 + \mathbf{2} \implies \mathbf{2} = \mathbf{42} - \mathbf{10} \cdot 4$$
$$\mathbf{10} = \mathbf{2} \cdot 5 + 0 \quad \text{(Stop)}$$

Step 2: Backward Path

$$2 = \mathbf{42} - \mathbf{10} \cdot 4$$
$$2 = \mathbf{42} - (52 - \mathbf{42} \cdot 1) \cdot 4$$
$$2 = \mathbf{42} \cdot 5 - 52 \cdot 4$$
$$2 = (198 - 52 \cdot 3) \cdot 5 - 52 \cdot 4$$
$$2 = 198 \cdot 5 - 52 \cdot 15 - 52 \cdot 4$$
$$\mathbf{2} = \mathbf{198} \cdot (5) + \mathbf{52} \cdot (-19)$$

Bézout coefficients: $\mathbf{s} = -\mathbf{19}$ and $\mathbf{t} = \mathbf{5}$

Consequences of Bézout's Theorem

- If $a, b, c \in \mathbb{Z}^+$ s.t. $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

  Proof: Since $\gcd(a, b) = 1$, there are integers s, t with

  $$sa + tb = 1 \implies sac + tbc = c$$

  $a \mid sac$ clearly, and $a \mid tbc$ because $a \mid bc$. Thus $a \mid (sac + tbc)$, meaning $a \mid c$.
- If p is prime and $p \mid a_1 a_2 \ldots a_n$, then $p \mid a_i$ for some i.
- Prime factorization is unique.

  Proof Sketch: From two representations, Cancel all common primes until no primes are shared
- Let $m \in \mathbb{Z}^+$ and let $a, b, c \in \mathbb{Z}$. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then: $a \equiv b \pmod{m}$.

Consequences of Bézout's Theorem

- If $a, b, c \in \mathbb{Z}^+$ s.t. $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

  Proof: Since $\gcd(a, b) = 1$, there are integers s, t with

  $$sa + tb = 1 \implies sac + tbc = c$$

  $a \mid sac$ clearly, and $a \mid tbc$ because $a \mid bc$. Thus $a \mid (sac + tbc)$, meaning $a \mid c$.

- If p is prime and $p \mid a_1 a_2 \ldots a_n$, then $p \mid a_i$ for some i.
- Prime factorization is unique.

  Proof Sketch: From two representations, Cancel all common primes until no primes are shared

- Let $m \in \mathbb{Z}^+$ and let $a, b, c \in \mathbb{Z}$. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then: $a \equiv b \pmod{m}$.

Consequences of Bézout's Theorem

- If $a, b, c \in \mathbb{Z}^+$ s.t. $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.
  Proof: Since $\gcd(a, b) = 1$, there are integers s, t with

  $$sa + tb = 1 \implies sac + tbc = c$$

  $a \mid sac$ clearly, and $a \mid tbc$ because $a \mid bc$. Thus $a \mid (sac + tbc)$, meaning $a \mid c$.
- If p is prime and $p \mid a_1 a_2 \dots a_n$, then $p \mid a_i$ for some i.
- Prime factorization is unique.
  Proof Sketch: From two representations, Cancel all common primes until no primes are shared
- Let $m \in \mathbb{Z}^+$ and let $a, b, c \in \mathbb{Z}$. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then: $a \equiv b \pmod{m}$.

Consequences of Bézout's Theorem

- If $a, b, c \in \mathbb{Z}^+$ s.t. $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.
  Proof: Since $\gcd(a, b) = 1$, there are integers s, t with

  $$sa + tb = 1 \implies sac + tbc = c$$

  $a \mid sac$ clearly, and $a \mid tbc$ because $a \mid bc$. Thus $a \mid (sac + tbc)$, meaning $a \mid c$.
- If p is prime and $p \mid a_1 a_2 \ldots a_n$, then $p \mid a_i$ for some i.
- Prime factorization is unique.

  Proof Sketch: From two representations, Cancel all common primes until no primes are shared
- Let $m \in \mathbb{Z}^+$ and let $a, b, c \in \mathbb{Z}$. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then: $a \equiv b \pmod{m}$.

Consequences of Bézout's Theorem

- If $a, b, c \in \mathbb{Z}^+$ s.t. $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.
  Proof: Since $\gcd(a, b) = 1$, there are integers s, t with
  $$sa + tb = 1 \implies sac + tbc = c$$
  $a \mid sac$ clearly, and $a \mid tbc$ because $a \mid bc$. Thus $a \mid (sac + tbc)$, meaning $a \mid c$.
- If p is prime and $p \mid a_1 a_2 \ldots a_n$, then $p \mid a_i$ for some i.
- Prime factorization is unique.
  Proof Sketch: From two representations, Cancel all common primes until no primes are shared
- Let $m \in \mathbb{Z}^+$ and let $a, b, c \in \mathbb{Z}$. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then: $a \equiv b \pmod{m}$.

Consequences of Bézout's Theorem

- If $a, b, c \in \mathbb{Z}^+$ s.t. $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.
  Proof: Since $\gcd(a, b) = 1$, there are integers s, t with

  $$sa + tb = 1 \implies sac + tbc = c$$

  $a \mid sac$ clearly, and $a \mid tbc$ because $a \mid bc$. Thus $a \mid (sac + tbc)$, meaning $a \mid c$.
- If p is prime and $p \mid a_1 a_2 \ldots a_n$, then $p \mid a_i$ for some i.
- Prime factorization is unique.
  Proof Sketch: From two representations, Cancel all common primes until no primes are shared
- Let $m \in \mathbb{Z}^+$ and let $a, b, c \in \mathbb{Z}$. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then: $a \equiv b \pmod{m}$.

Euclidean Algorithm: An Example

Use the Euclidean algorithm to find $\gcd(123, 277)$.

$$\begin{aligned}277 &= 123 \cdot 2 + 31,\\ 123 &= 31 \cdot 3 + 30,\\ 31 &= 30 \cdot 1 + 1,\\ 30 &= 1 \cdot 30 + 0.\end{aligned}$$

The last nonzero remainder is 1. Therefore,

$$\boxed{\gcd(123, 277) = 1}.$$

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

$\mathcal{Q}$. How many zeros are there at the end of 100!?

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

$\mathcal{Q}$. How many zeros are there at the end of 100!?

$\mathcal{A}$. Count the factors of 5: $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = 24$. Hence, $\boxed{24}$ zeros.

Prime Factorization: Exercises

$\mathfrak{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

$\mathfrak{Q}$. How many zeros are there at the end of 100!?

$\mathcal{A}$. Count the factors of 5: $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = 24$. Hence, $\boxed{24}$ zeros.

$\mathfrak{Q}$. Find $\gcd(3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21})$.

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

$\mathcal{Q}$. How many zeros are there at the end of 100!?

$\mathcal{A}$. Count the factors of 5: $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = 24$. Hence, $\boxed{24}$ zeros.

$\mathcal{Q}$. Find $\gcd(3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21})$.

$\mathcal{A}$. The two numbers have no common prime factor, so $\boxed{1}$.

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

$\mathcal{Q}$. How many zeros are there at the end of 100!?

$\mathcal{A}$. Count the factors of 5: $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = 24$. Hence, $\boxed{24}$ zeros.

$\mathcal{Q}$. Find $\gcd(3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21})$.

$\mathcal{A}$. The two numbers have no common prime factor, so $\boxed{1}$.

$\mathcal{Q}$. How many divisions are required to find $\gcd(34, 55)$ using the Euclidean algorithm?

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

$\mathcal{Q}$. How many zeros are there at the end of 100!?

$\mathcal{A}$. Count the factors of 5: $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = 24$. Hence, $\boxed{24}$ zeros.

$\mathcal{Q}$. Find $\gcd(3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21})$.

$\mathcal{A}$. The two numbers have no common prime factor, so $\boxed{1}$.

$\mathcal{Q}$. How many divisions are required to find $\gcd(34, 55)$ using the Euclidean algorithm?

$\mathcal{A}$. 8 divisions

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

$\mathcal{Q}$. How many zeros are there at the end of 100!?

$\mathcal{A}$. Count the factors of 5: $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = 24$. Hence, $\boxed{24}$ zeros.

$\mathcal{Q}$. Find $\gcd(3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21})$.

$\mathcal{A}$. The two numbers have no common prime factor, so $\boxed{1}$.

$\mathcal{Q}$. How many divisions are required to find $\gcd(34, 55)$ using the Euclidean algorithm?

$\mathcal{A}$. 8 divisions

$\mathcal{Q}$. Prove or disprove that $n^2 - 79n + 1601$ is prime whenever n is a positive integer.

Prime Factorization: Exercises

$\mathcal{Q}$. Find the prime factorization of 10!.

$\mathcal{A}$. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

$\mathcal{Q}$. How many zeros are there at the end of 100!?

$\mathcal{A}$. Count the factors of 5: $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = 24$. Hence, $\boxed{24}$ zeros.

$\mathcal{Q}$. Find $\gcd(3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21})$.

$\mathcal{A}$. The two numbers have no common prime factor, so $\boxed{1}$.

$\mathcal{Q}$. How many divisions are required to find $\gcd(34, 55)$ using the Euclidean algorithm?

$\mathcal{A}$. 8 divisions

$\mathcal{Q}$. Prove or disprove that $n^2 - 79n + 1601$ is prime whenever n is a positive integer.

$\mathcal{A}$. False. Take $n = 80$: $80^2 - 79(80) + 1601 = 1681 = 41^2$.

Prime Factorization: Exercises

Euler ϕ-function

For a positive integer *n*, the **Euler ϕ-function** is defined to be the number of positive integers less than or equal to n that are relatively prime to n. For instance, $\phi(6) = 2$.

Prime Factorization: Exercises

Euler ϕ-function

For a positive integer n, the **Euler ϕ-function** is defined to be the number of positive integers less than or equal to n that are relatively prime to n. For instance, $\phi(6) = 2$.

$\mathfrak{Q}$. What is $\phi(p)$ where p is a prime?

Prime Factorization: Exercises

Euler ϕ-function

For a positive integer n, the **Euler ϕ-function** is defined to be the number of positive integers less than or equal to n that are relatively prime to n. For instance, $\phi(6) = 2$.

$\mathfrak{Q}$. What is $\phi(p)$ where p is a prime?

$\mathcal{A}$. $\phi(p) = p - 1$.

Prime Factorization: Exercises

Euler ϕ-function

For a positive integer n, the **Euler ϕ-function** is defined to be the number of positive integers less than or equal to n that are relatively prime to n. For instance, $\phi(6) = 2$.

$\mathfrak{Q}$. What is $\phi(p)$ where p is a prime?

$\mathcal{A}$. $\phi(p) = p - 1$.

$\mathfrak{Q}$. What is $\phi(p^k)$ where p is a prime?

Prime Factorization: Exercises

Euler ϕ-function

For a positive integer n, the **Euler ϕ-function** is defined to be the number of positive integers less than or equal to n that are relatively prime to n. For instance, $\phi(6) = 2$.

$\mathfrak{Q}$. What is $\phi(p)$ where p is a prime?

$\mathcal{A}$. $\phi(p) = p - 1$.

$\mathfrak{Q}$. What is $\phi(p^k)$ where p is a prime?

$\mathcal{A}$. $\phi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$.

Prime Factorization: Exercises

Euler ϕ-function

For a positive integer n, the **Euler ϕ-function** is defined to be the number of positive integers less than or equal to n that are relatively prime to n. For instance, $\phi(6) = 2$.

$\mathfrak{Q}$. What is $\phi(p)$ where p is a prime?

$\mathcal{A}$. $\phi(p) = p - 1$.

$\mathfrak{Q}$. What is $\phi(p^k)$ where p is a prime?

$\mathcal{A}$. $\phi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$.

$\mathfrak{Q}$. Show that $\phi(pq) = \phi(p)\phi(q)$ where p and q are relatively prime.

Prime Factorization: Exercises

Euler ϕ-function

For a positive integer n, the **Euler ϕ-function** is defined to be the number of positive integers less than or equal to n that are relatively prime to n. For instance, $\phi(6) = 2$.

$\mathcal{Q}$. What is $\phi(p)$ where p is a prime?

$\mathcal{A}$. $\phi(p) = p - 1$.

$\mathcal{Q}$. What is $\phi(p^k)$ where p is a prime?

$\mathcal{A}$. $\phi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$.

$\mathcal{Q}$. Show that $\phi(pq) = \phi(p)\phi(q)$ where p and q are relatively prime.

$\mathcal{Q}$. What is $\phi(10)$?

Prime Factorization: Exercises

Euler ϕ-function

For a positive integer n, the **Euler ϕ-function** is defined to be the number of positive integers less than or equal to n that are relatively prime to n. For instance, $\phi(6) = 2$.

$\mathcal{Q}$. What is $\phi(p)$ where p is a prime?

$\mathcal{A}$. $\phi(p) = p - 1$.

$\mathcal{Q}$. What is $\phi(p^k)$ where p is a prime?

$\mathcal{A}$. $\phi(p^k) = p^k - p^{k-1} = p^{k-1}(p-1)$.

$\mathcal{Q}$. Show that $\phi(pq) = \phi(p)\phi(q)$ where p and q are relatively prime.

$\mathcal{Q}$. What is $\phi(10)$?

$\mathcal{A}$. $\phi(10) = 4$.

Thank You for your attention!

Any questions?